StoReD: Stochastic Readout of Distributional Memory

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Abstract

External memory offers a practical way to improve frozen LLM agents without updating their parameters. Existing work has focused primarily on what experience should be stored and retrieved, while largely treating the retrieved context as fixed during execution. When a retrieved entry contains multiple valid suggestions, existing readers handle this uncertainty at two sub-optimal extremes: a static Maximum a Posteriori (MAP) reader prematurely commits to the single most frequent guidance for all retries, forcing subsequent attempts to inherit the same mistaken assumption; conversely, a List reader injects all candidates into one prompt, forcing the frozen agent to dynamically arbitrate competing instructions. To address this, we introduce **StoReD**, a training-free, attempt-wise readout mechanism that makes a fresh guidance decision for each retry. Offline, a stronger teacher distills experience into a frequency-weighted prior over plausible guidelines. At test time, retrieval remains stable, but our reader samples one coherent instruction per attempt, drawing a fresh guidance upon failure without test-time teacher calls. Theoretically, we prove that independent semantic redrawing weakly dominates static commitment in minimising multi-attempt failure probabilities. Empirically, on ALFWorld, teacher-distilled memory first lifts single-attempt success from 35.0 to 49.0 over the memory-free student ($+14.0$ points at success@1), and StoReD preserves this strong single-attempt benefit while dramatically improving retry complementarity, outperforming a static MAP reader by a $+12.0$ absolute margin in success@8. Crucially, a matched sample-once control confirms this gain stems specifically from breaking the joint dependence of failures, elevating an unguided 4B student from roughly half of the 32B teacher's baseline performance to an impressive 93% of its multi-attempt success.

1 Introduction

External memory offers a practical route to stronger language agents when model parameters cannot be updated. A frozen agent can retrieve records, reflections, skills, or distilled instructions and condition its next action on them (Park et al. 2023; Shinn et al. 2023; Wang et al. 2023a; Packer et al. 2023). At the same time, modern agent systems increasingly rely on *loop engineering*: they observe environment or verifier feedback, replan, reflect, and retry (Yao et al. 2023; Shinn et al. 2023). Success@k (Chen et al. 2021) is therefore an operational system objective rather than merely an evaluation convenience. A useful memory should improve the first attempt and make later attempts use the retry budget productively.

Most memory research concentrates on the write and retrieval sides: what experience to preserve? It is framed as a constrained retention problem for long-horizon agents (Kang et al. 2026b): how to consolidate it, with the retention-versus-consolidation choice shown to be budget-dependent (Kang et al. 2026a), and which entry to retrieve. We study the next decision in the loop: once an entry has been retrieved, how should its guidance be consumed across repeated attempts?

A retrieved entry may support several valid ways to proceed. Existing readers place this decision at two opposite extremes. A static MAP reader selects the most frequent guidance and reuses it throughout the retry sequence. If that mode is mismatched to the current state, every retry inherits the same high-level assumption. A List reader retains the alternatives but concatenates them into every prompt, forcing the frozen executor to select among and reconcile competing instructions inside a single attempt. *MAP resolves memory uncertainty too early; List resolves it too late.*

We introduce **StoReD**, which resolves the choice at the attempt boundary. Offline, we roll out the frozen student at the states it visits and draw multiple guidance samples from a stronger black-box teacher. Semantic abstraction and clustering convert these samples into a compact distributional memory: a support of teacher-endorsed guidance together with empirical recommendation frequencies. At test time, the teacher is absent. StoReD samples one coherent instruction for the current attempt, keeps it stable within that attempt, and draws again after failure. The mode remains the most likely instruction, while lower-frequency teacher-supported alternatives continue to receive opportunities across retries. The method is designed for agent loops in which an environment or a reliable verifier can accept a successful rollout or trigger another reset attempt; with a single attempt, it reduces to one draw from the stored distribution.

The key object is not only a distribution over guidance in one attempt, but a joint readout law across attempts. A complete reader specifies the retained candidate support, the one-attempt marginal over that support, and the coupling of guidance choices across retries. MAP collapses the marginal

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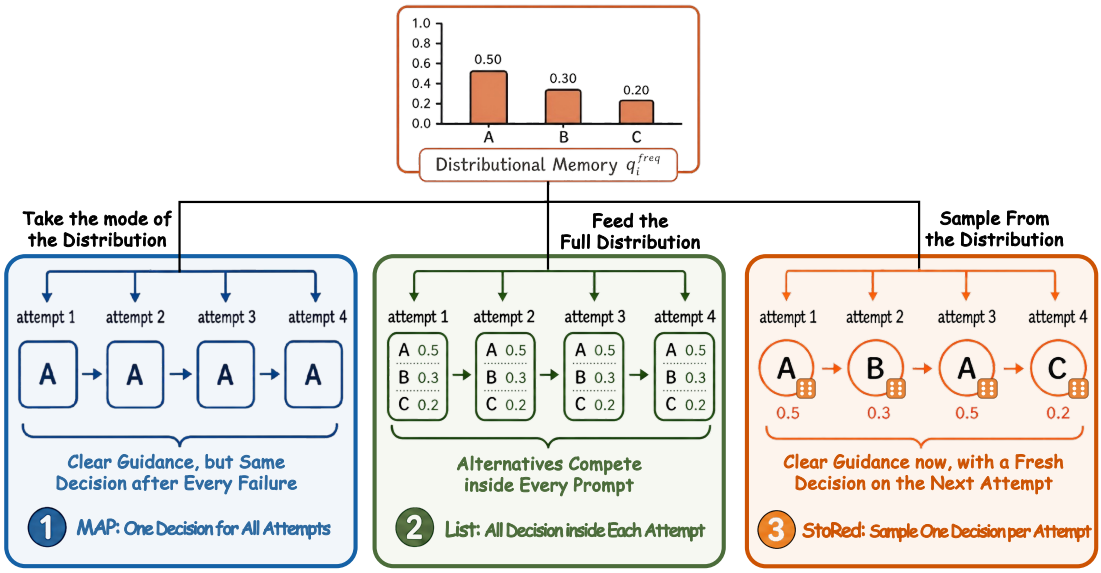


Figure 1: Three readouts of the same retrieved entry. MAP makes one guidance decision for the entire retry sequence. List transfers the unresolved choice to the frozen executor. StoReD makes one coherent guidance decision at each attempt boundary, preserving within-attempt consistency while allowing later attempts to reconsider alternatives already stored in memory.

to its mode. Sample-Once retains the same teacher-frequency marginal as StoReD but shares one random draw across all attempts. StoReD independently redraws from that marginal at every attempt, preserving a coherent plan within a rollout while turning retries into complementary trials.

The gains appear at both timescales. With Qwen3-32B memory on ALFWorld, StoReD raises success@1 from 35.0 to 49.0, including a 2.5-point gain over MAP. With independently distilled GLM-5.2 memory, it raises success@1 from 35.0 to 57.3 and exceeds MAP by 6.0 points. At eight attempts, Qwen-memory StoReD reaches 79.9, compared with 67.9 for MAP and 73.2 for Sample-Once averaged over five independent commitment maps. Under GLM memory, StoReD reaches 82.1, compared with 72.4 for MAP and a five-map Sample-Once mean of 80.3. A denser 32-attempt verification with Qwen memory increases the redraw advantage over Sample-Once to 9.6 points.

A diagnostic probe explains why storing the distribution matters. Across 70 probed entries, the teacher mode is the candidate the frozen student executes best in 60% of cases, while a non-mode candidate is better in the remaining 40%. Teacher frequency ranks what the teacher tends to recommend, not what the weaker student can most reliably execute; it is a useful prior rather than a calibrated student-utility ranking. The three readers expose both sides of this distinction. Uniform redrawing improves over MAP by preserving and revisiting the teacher-supported tail, while frequency-weighted StoReD further improves over uniform in all four teacher–environment panels. The resulting rule is to prioritise teacher consensus without permanently excluding lower-frequency alternatives.

Our analysis treats each completed memory entry as a finite empirical bank. The practical collection budget is $m = 8$ teacher calls per visited state, but after states are aggregated and retained counts are formed, entry i contains an effective bank of N_i sample slots. The theory separates this offline bank size from the online retry budget k: under a canonical exchangeable-slot model, finite-bank redrawing lies between Sample-Once and population redrawing, bank-induced dependence decays as $1/N_i$, and an occupancy law characterises how completely the fixed bank is explored. These are design predictions for varying the effective bank size; the experiments directly test the fixed-bank coupling implication rather than the monotonicity in N_i.

The two ALFWorld memories further connect stored diversity to the opportunity available to readout. Qwen3-32B memory retains 3.4 semantic candidates per non-empty entry, compared with 1.7 for GLM-5.2, and exhibits the larger relative MAP-to-StoReD gain at success@8 (17.7% versus 13.4%). Candidate count is not itself the governing quantity: retry headroom arises when the retained alternatives induce heterogeneous student success. We therefore treat the cross-teacher ordering as an empirical association consistent with, rather than a causal test of, the theory.

This paper makes three contributions:

1. **Attempt-wise StoReD readout.** We identify post-retrieval guidance selection as a distinct memory layer and introduce a training-free reader that gives the frozen agent one coherent instruction within an attempt and reopens the choice after failure.
2. **Distributional memory as support, marginal, and coupling.** We separate which teacher-supported alternatives are stored, how they are weighted in one attempt, and how their draws are coupled across retries. A utility probe shows why the teacher-frequency prior and its lower-frequency tail are both valuable to a frozen student.
3. **Finite-bank theory and matched mechanism evidence.**

We model a completed memory entry as a random empirical measure with effective slot count N_i and derive a monotone finite-bank hierarchy, a $1/N_i$ residual-dependence law, and a two-budget exposure law linking offline bank size to online retries. An allocation–coupling decomposition maps these quantities to MAP, Sample-Once, and StoReD. Five-map Sample-Once controls under two independently constructed teacher memories isolate the coupling term, while two interactive environments test the complete reader.

2 Related Work

External memory for agents. Language-agent memories store episodic records, reflections, reusable skills, manuals, and trajectories (Park et al. 2023; Shinn et al. 2023; Wang et al. 2023a; Packer et al. 2023; Zhou et al. 2024; Kagaya et al. 2024; Sarukkai, Xie, and Fatahalian 2025). Memory systems further study extraction, consolidation, conflict resolution, retention, and retrieval (Chhikara et al. 2025; Li et al. 2025). These choices mainly concern which content reaches the agent. Our intervention begins after a multi-candidate entry has been retrieved and asks how that fixed content should be consumed across retries.

Teacher-distilled guidance. A strong model can convert experience into reusable guidelines, manuals, or insights for a weaker executor (Fu et al. 2024; Chen et al. 2024; Zhao et al. 2024). Querying the teacher at learner-visited states follows the covariate-shift motivation of DAgger and knowledge distillation (Ross, Gordon, and Bagnell 2011; Hinton, Vinyals, and Dean 2015; Gu et al. 2024; Agarwal et al. 2024). Our write pipeline is compatible with this family—including training-free and inference-time agent distillation (Qiu et al. 2025; Sarukkai et al. 2025) and sub-goal-level distillation for small agents (Hashemzadeh et al. 2024; Nekoei et al. 2025), but our contribution is the read semantics of repeated teacher outputs rather than a new parameter-training objective (Feng et al. 2025).

Uncertainty-aware memory and adaptation. Recent work stores multiple beliefs, alters output distributions using retrieved experience, or applies Bayesian and bandit mechanisms (Thompson 1933; Russo et al. 2018) to memory acquisition, retrieval, and procedure selection (Liao et al. 2026; Li et al. 2026; Wu et al. 2026; Xu et al. 2026; Forouzandeh et al. 2025; Yang et al. 2024b; Yan et al. 2026). We isolate a narrower setting: the retrieved entry and its candidate support are fixed, and the reader determines when uncertainty is resolved. StoReD does not estimate online utility, update the memory from rewards, or train a readout policy.

Inference-time diversity. Self-consistency, best-of-N, prompt perturbation, and varied exemplars improve coverage by changing reasoning traces or prompts across samples (Wang et al. 2023b, 2025; Michelman, Baratalipour, and Abueg 2025; Di et al. 2025; Dalal et al. 2025; Brown et al. 2024). StoReD places this diversity decision at a different layer: the alternatives are state-local semantic guidance already stored in memory, and the reader controls whether one memory decision is shared or reopened across attempts. Each attempt still receives one coherent teacher-produced context at a fixed decoder temperature; the model neither invents prompts online nor arbitrates a list of competing instructions within the same attempt. The contribution is therefore the memory-readout formulation and matched coupling comparison, not the generic observation that repeated sampling can improve coverage.

3 Problem Setup: Guidance Distributions and Retry Coupling

Frozen student and retry objective. A task $x \in \mathcal{X}$ fixes an initial state $s_0(x)$; within a rollout, s denotes the current state or interaction history. The frozen student selects a single environment action according to

$$A \sim \pi_{\mathrm{S}}(\cdot \mid s, z), \tag{1}$$

where z is an optional natural-language guidance candidate; the student parameters stay fixed during both memory construction and evaluation. Each task is evaluated through at most k fresh rollouts from $s_0(x)$. After rollout a, the environment or a reliable verifier returns $Y_a \in \{0, 1\}$; a success is accepted immediately, and a failure triggers another reset rollout with a fresh attempt-level readout. Attempts are never aggregated:

$$\mathrm{success@}\, k = \Pr\left(\max_{1 \le a \le k} Y_a = 1\right). \tag{2}$$

A fixed retriever, shared by all methods, maps the state to an entry $i = r(s)$; our question begins after entry i has been retrieved.

Teacher guidance as a distribution. At a student-visited state, the black-box teacher produces $G \sim \pi_{\mathrm{T}}(\cdot \mid s)$, and the operator ϕ abstracts one raw response into an instance-agnostic candidate $Z = \phi(G)$; entry-level semantic clustering subsequently merges equivalent realisations. A realisation z is a complete instruction supplied to the frozen student—neither an environment action nor an entire memory entry. With ν_i the distribution of states represented by entry i, the teacher induces the entry-level guidance distribution

$$p_i(z) = \mathbb{E}_{s \sim \nu_i}[\Pr(\phi(G) = z \mid s)]. \tag{3}$$

Repeated black-box samples estimate both the support of plausible guidance and how often the teacher produces each candidate. The memory retains a finite set $\mathcal{Z}_i$; to distinguish support truncation from frequency estimation, let $\bar{p}_i$ denote p_i renormalised on $\mathcal{Z}_i$,

$$\bar{p}_i(z) = \frac{p_i(z)}{\sum_{z' \in \mathcal{Z}_i} p_i(z')}, \qquad z \in \mathcal{Z}_i, \tag{4}$$

which the stored $q_i \in \Delta(\mathcal{Z}_i)$ estimates. Sampling guidance and executing it through the frozen student induces

$$\pi_{q_i}(a \mid s, i) = \sum_{z \in \mathcal{Z}_i} q_i(z)\, \pi_{\mathrm{S}}(a \mid s, z), \tag{5}$$

and $\pi_{\mathrm{T} \to \mathrm{S}}$ denotes the same mixture under $\bar{p}_i$; the two coincide when $q_i = \bar{p}_i$. The target is therefore a teacher-*guided* frozen student, not the teacher acting directly; any remaining gap reflects the student's ability to execute the supplied guidance.

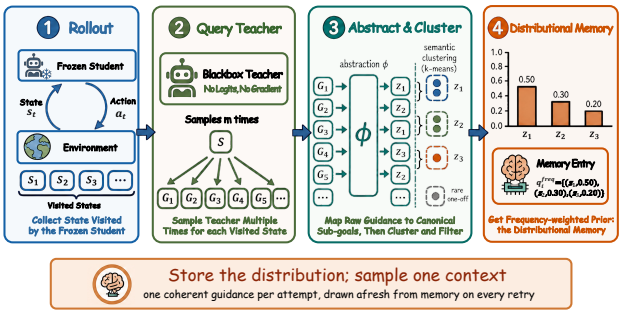


Figure 2: Building and reading the distributional memory. (1) The frozen student explores; its visited states are collected. (2) A black-box teacher (no logits, no gradients) is sampled m times per state. (3) Responses are abstracted into canonical sub-goals, clustered, and filtered of one-off noise. (4) Each entry stores the retained candidates with their empirical frequencies. At test time, the readout samples one guidance per attempt and draws afresh on every retry.

A readout specifies support, marginal, and coupling. The support and one-attempt marginal do not fully specify a reader under retries; the reader must also define the joint law over the guidance exposed across attempts. Three readers illustrate the distinction. MAP collapses the marginal to the teacher-frequency mode and shares it:

$$z_i^\star = \arg\max_z q_i(z), \qquad Z_a = z_i^\star \quad \text{for all } a. \tag{6}$$

Sample-Once preserves q_i but shares one random draw:

$$Z \sim q_i, \qquad Z_a = Z \quad \text{for all } a. \tag{7}$$

StoReD independently redraws from the same marginal:

$$Z_a \overset{\text{iid}}{\sim} q_i, \qquad a = 1, \dots, k. \tag{8}$$

Sample-Once and StoReD thus share the one-attempt marginal and differ only in cross-attempt dependence; at $k = 1$ the two procedures coincide. A List reader lies outside this latent-choice family: a deterministic composite prompt containing all retained candidates, whose arbitration is left to the frozen executor. State-dependence is handled by attempt-level readout maps: the entry marginals induce a distribution ρ_x over complete maps, a draw $M_a \sim \rho_x$ assigning each potentially queried entry a candidate $M_a(i) \sim q_i$, with only trajectory-encountered coordinates revealed and the selected guidance free to change later states and retrievals. Sample-Once shares one map across attempts; StoReD draws $M_1, \dots, M_k \overset{\text{iid}}{\sim} \rho_x$. The comparison therefore does not require different attempts to visit the same states or retrieve the same entries. The marginal determines the behaviour induced in one attempt; the joint law determines whether later attempts repeat the same memory decision or reconsider the alternatives.

4 StoReD: Stochastic Readout of Distributional Memory

Monte Carlo memory distillation. Figure 2 summarises the construction. We roll out the frozen student and collect its visited states. At each state s_n we issue m independent decoding calls to the same black-box teacher (fixed prompt and decoding configuration), modelled as conditionally i.i.d. given the state, and abstract each response, $z_{n,j} = \phi(G_{n,j})$. Semantically equivalent candidates within an entry are clustered and one-off outputs filtered. With $c_i(z)$ the retained count of canonical candidate z and

$$N_i = \sum_{z \in \mathcal{Z}_i} c_i(z) \tag{9}$$

the entry's effective number of retained slots, the retained counts define the default marginal

$$q_i^{\text{freq}}(z) = c_i(z)/N_i, \tag{10}$$

equivalently uniform selection among the N_i slots, with repeated candidates occupying repeated slots. This empirical distribution is a black-box distillation prior: it preserves teacher consensus while retaining repeatedly endorsed alternatives, and requires no teacher logits, student gradients, or outcome labels.

Attempt-wise stochastic readout and uniform ablation. At test time the teacher is absent. When entry i is retrieved on attempt a, StoReD samples $Z_{a,i} \sim q_i^{\text{freq}}$ and supplies the realised candidate as one coherent instruction; the draw is reused if the entry recurs within that attempt and redrawn for the next attempt—stochasticity lives at the attempt boundary, not inside a single execution trajectory. To isolate the value of the empirical weights, we also evaluate the uniform marginal $q_i^{\text{unif}}(z) = 1/|\mathcal{Z}_i|$ with the retained candidates, retrieval, prompts, student, and attempt-wise redrawing unchanged, separating the value of the stored support from the allocation gain of teacher frequency.

5 Finite-Bank Theory of Distributional Readout

Distributional memory has two finite budgets: the pipeline spends m teacher calls per visited state, and a completed entry contains N_i retained empirical slots after aggregation, abstraction, clustering, and filtering; frequency-weighted readout from q_i is exactly uniform selection among those slots. We state conclusions here and defer all proofs and auxiliary results to the supplementary material. To isolate the sampling effect, consider a canonical entry with N exchangeable slots $G_1, \dots, G_N \overset{\text{iid}}{\sim} P_x$, utilities $U_j = u_x(G_j)$ with mean μ_x and variance σ_x^2, and bank average $\overline{U}_N$. The all-failure probabilities of sharing one guidance, redrawing from the finite bank, and redrawing from the population are

$$F_k^{\text{once}} = \mathbb{E}[(1 - U_1)^k], \quad F_{N,k}^{\text{bank}} = \mathbb{E}[(1 - \overline{U}_N)^k], \tag{11}$$

$$F_k^{\text{pop}} = (1 - \mu_x)^k. \tag{12}$$

Theorem 1 (Finite-bank hierarchy) *For every fixed $k \geq 2$,*

$$F_k^{\text{pop}} \leq F_{N+1,k}^{\text{bank}} \leq F_{N,k}^{\text{bank}} \leq F_k^{\text{once}}, F_{N,k}^{\text{bank}} \xrightarrow[N\to\infty]{} F_k^{\text{pop}}, \tag{13}$$

with the finite-bank expansion

$$F_{N,k}^{\text{bank}} = (1-\mu_x)^k + \frac{k(k-1)}{2N}(1-\mu_x)^{k-2}\sigma_x^2 + O(N^{-2}), \tag{14}$$

and, under independent slot selection from the same random bank, $\mathrm{Cov}(Y_a, Y_b) = \sigma_x^2/N$ *for* $a \neq b$*, so* $\mathrm{Var}(\sum_{a=1}^k Y_a) = k\mu_x(1-\mu_x) + \frac{k(k-1)}{N}\sigma_x^2$.

Finite-bank redrawing thus lies between Sample-Once and ideal population redrawing in convex order (Rothschild and Stiglitz 1970); increasing the effective bank size monotonically contracts the shared-guidance component of failure, whose leading penalty and inter-attempt covariance both decay as $1/N$. The theorem is a design prediction for changing the bank size; our experiments keep each constructed bank fixed and directly test the same-bank coupling implication.

Proposition 1 (Offline-discovery–online-exploration law) *k independent readouts from a fixed bank of size N expose $\mathbb{E}[D_{N,k}] = N[1-(1-1/N)^k]$ distinct slots. Under the exchangeable-slot model, a useful guidance region of population mass β is exposed at least once with probability*

$$C_{N,k}(\beta) = 1 - \sum_{n=0}^{N} \binom{N}{n} \beta^n (1-\beta)^{N-n} \left(1 - \tfrac{n}{N}\right)^k, \tag{15}$$

with $C_{N,1}(\beta) = \beta$ and $\lim_{k\to\infty} C_{N,k}(\beta) = 1-(1-\beta)^N$.

The offline bank determines which alternatives can ever be exposed; the retry budget determines how completely that fixed bank is explored, and additional retries cannot recover guidance absent from the completed entry. The full occupancy distribution and rare-tail limit appear in the supplementary material.

Task-level reader decomposition. With $M \sim \rho_x$ the attempt-level readout map of Section 3, $U_x = u_x(M)$, and $M^\star$ the deterministic MAP map,

$$S_k^{\text{redraw}} - S_k^{\text{MAP}} = \underbrace{S_k^{\text{once}} - S_k^{\text{MAP}}}_{\text{distributional marginal}} + \underbrace{S_k^{\text{redraw}} - S_k^{\text{once}}}_{\text{retry coupling}}, \tag{16}$$

where the same-marginal coupling term satisfies $\Delta_2(x) = \mathrm{Var}(U_x)$ and $0 \le \Delta_k(x) \le \frac{k(k-1)}{2}\mathrm{Var}(U_x)$—exactly the quantity measured by Sample-Once versus StoReD. For one retained support of size d, frequency improves over uniform allocation precisely when teacher propensity and frozen-student utility are positively aligned:

$$\sum_z q(z)\,u_x(z) - \frac{1}{d}\sum_z u_x(z) = d\ \mathrm{Cov}_{\mathrm{Unif}}\big(q(Z), u_x(Z)\big). \tag{17}$$

The convex-order proof, occupancy distribution, rare-tail asymptotics, task-level Jensen bounds, and behavioural contraction through the frozen student are given in the supplementary material.

6 Experiments

We test three claims: distributional teacher memory improves an individual attempt; attempt-wise redrawing turns retries into complementary trials; and empirical teacher frequency improves allocation over the retained support. We also relate the diversity of two independently constructed teacher memories to the readout gain they expose. Pairwise intervals use a task-level paired bootstrap (Efron and Tibshirani 1994) (B=10,000, two-sided 95% CIs).

Setup. ALFWorld (Shridhar et al. 2021): frozen Qwen3-4B student (Yang et al. 2025) (thinking disabled, temperature 0.6) in a ReAct loop (Yao et al. 2023); 134 `valid_unseen` tasks, n=8 attempts. The Qwen3-32B teacher is queried m=8 times per student-visited state; 107,826 states from 3,553 training games compress into 9,512 entries (38 MB) retaining 3.4 candidates per entry and 0.94 of the teacher mass, with responses abstracted into instance-agnostic subgoals. The GLM-5.2 memory uses the same pipeline and retains 1.7 candidates per entry. WebShop (Yao et al. 2022): human-written 500-task split, n=4; the student is Qwen2.5-7B-Instruct (Yang et al. 2024a), fixed before any reader experiment after Qwen3-4B proved near the task floor; the memory stores literal query/action suggestions ($\phi(y)$=y). Within each benchmark and teacher panel, all readers share the bank, keys, retrieval, support, student, scaffold, temperature, and budget.

ALFWorld: stronger attempts and complementary retries. Table 1 reports the main comparison. Distributional memory lifts the first attempt substantially, and StoReD lifts it further: success@1 rises 35.0 → 49.0 under Qwen memory (+2.5 over MAP) and 35.0 → 57.3 under GLM memory (+6.0 over MAP), sampling lets teacher-supported alternatives match the exact state rather than always receiving the global mode. Retries amplify the difference. With Qwen memory, StoReD reaches 79.9 success@8 versus 67.9 (MAP) and 73.2 (Sample-Once): 5.3 points for replacing the mode by the stored distribution, another 6.7 for reopening it across attempts. With GLM memory the terms are 7.9 and 1.8 (80.3 and 82.1). Redrawing thus improves over a fixed draw under both memories, with the split between marginal diversification and retry coupling differing across them; at 32 attempts the Qwen-memory redraw gain grows to 9.6 points (p<.001).

Matched coupling control. Sample-Once and StoReD expose the same teacher-frequency marginal, one instruction at a time; they differ only in whether the sampled memory map is shared or redrawn. Five independent commitment maps obtain 73.2 ± 0.3 success@8 under Qwen memory (range 72.8–73.5) and 80.3 ± 0.8 under GLM memory (range 79.1–81.2); StoReD exceeds every map (+6.4–7.1 and +0.9–3.0 respectively). At success@1 the two readers coincide by construction, so Table 1 reports the shared measurement in both rows; coupling is a multi-attempt property, identified by the success@k comparison.

Teacher frequency is useful without being a student-utility model. We evaluate a fixed set of 70 anchor entries; the selection protocol and tie handling are reported in

ALFWorld system	s@1	s@8
Memory from a Qwen3-32B teacher		
BASE[§]	35.0	58.2
MAP (mode fixed)	46.5	67.9
LIST (all in one prompt)	46.7	67.9
SAMPLE-ONCE (frequency draw fixed)[‡]	49.0	73.2
UNIFORM (redraw)	47.9	79.1
STORED (frequency redraw)	**49.0**	**79.9**
Teacher-as-agent (32B)	64.4	85.8
Memory from a GLM-5.2 teacher		
BASE[§]	35.0	58.2
MAP (mode fixed)	51.3	72.4
LIST (all in one prompt)	53.3	73.9
SAMPLE-ONCE (frequency draw fixed)[‡]	57.3	80.3
UNIFORM (redraw)	56.1	79.5
STORED (frequency redraw)	**57.3**	**82.1**
Teacher-as-agent (GLM-5.2)[†]	91.0	—

Table 1: Matched ALFWorld comparison on `valid_unseen` (134 tasks, $n = 8$). With Qwen memory, STORED improves over MAP by 2.5 points at success@1 and 11.9 points at success@8 (95% CI $[+3.7, +20.1]$, $p = .005$ for success@8). With GLM memory, the corresponding gains are 6.0 and 9.7 points. Uniform uses the identical retained support and attempt-wise redrawing while flattening teacher frequencies. [§]BASE is memory-free; one shared run is reported in both panels. [‡]At success@8, the mean over five commitment maps (std 0.3 Qwen, 0.8 GLM); at success@1, Sample-Once and STORED coincide by construction, so the shared value is reported. [†]Single rollout ($n = 1$), not protocol-matched to the reader rows.

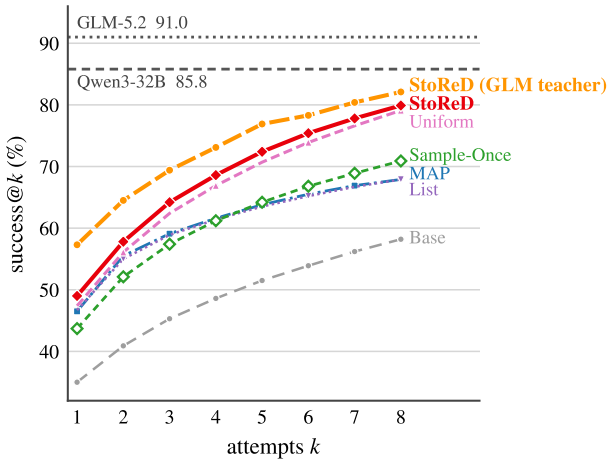


Figure 3: Success as the retry budget grows. The plotted same-marginal comparison uses Qwen memory: the gap between Sample-Once and STORED widens over the evaluated retry horizon. Table 1 additionally replicates the fixed-marginal comparison with GLM memory at success@8.

the supplementary material. For each retained candidate, we estimate student utility $\hat{u}(z)$ from $R = 32$ independent rollouts. The teacher mode attains the highest estimated utility in 60% of entries, while a non-mode candidate is better in the remaining 40%. The within-entry rank correlation between recommendation frequency and student utility is -0.14 (95% CI $[-0.34, +0.07]$). The mode is therefore often an effective default, but the frequency ordering does not calibrate how the frozen student executes all alternatives.

Table 2 shows the largest mode-to-best utility gap in the probe. The teacher recommends candidate A 81.2% of the time, yet the student succeeds with it only 35% of the time; two tail candidates are executed perfectly. STORED does not estimate or optimise these utilities. It simply retains the teacher-supported alternatives and repeatedly reopens the choice, allowing high-utility tail candidates to receive opportunities without abandoning the teacher prior.

Teacher frequency improves allocation over the same support. Frequency-weighted STORED is the strongest distributional reader in every teacher–environment panel. Relative to uniform redrawing over exactly the same retained candidates, it improves success@k by 0.8 points on Qwen ALFWorld, 2.6 on GLM ALFWorld, 0.8 on DeepSeek WebShop, and 1.3 on GLM WebShop. The repeated ordering shows that recommendation frequency carries useful allocation information beyond identifying the support. The strong uniform results reveal the complementary source of value: retaining the distributional tail gives the frozen student access to alternatives that MAP permanently discards.

Memory diversity and observed readout gain. The Qwen memory retains twice as many semantic candidates as the GLM memory (3.4 versus 1.7 per non-empty entry) and also exhibits the larger relative MAP-to-STORED improvement at success@8 (17.7% versus 13.4%). This ordering is consistent with the variance term, as a richer support can create more readout headroom when its candidates induce different student behaviours. Candidate count is not itself the variance, however, and the two teacher memories also differ in content and quality. We therefore treat this cross-teacher pattern as an observational association rather than a causal test of the headroom mechanism.

WebShop: gains under literal guidance. Table 3 reports the WebShop results. STORED is the strongest reader under both teacher memories. With DeepSeek memory, it raises success@4 from 24.2 for Base and MAP to 29.3. With GLM memory, it raises success@4 from 24.2 for Base to 33.2, outperforming MAP by 8.7 points and List by 4.4 points. Under DeepSeek memory, MAP also falls below Base on score (0.394 versus 0.435) and only matches it at success@4: committing every retry to one literal string can hurt, while the distributional readers recover and exceed Base under the same memory support.

The smaller tier separation relative to ALFWorld follows the memory representation. ALFWorld stores reusable semantic subgoals such as locating a sink or cleaning a held object. WebShop stores literal product names, attribute or-

candidate	w (teacher recommendation frequency)	$\hat{u}$ (student utility, R=32)
A: `go to the next unsearched place`	0.812	0.35
B: `go to the sink to clean the spatula`	0.100	**1.00**
C: `go to the sink to clean the dishsponge`	0.088	**1.00**

Table 2: An illustrative anchor selected as the largest mode-to-best gap in the 70-entry probe. The teacher mode receives 81.2% of recommendations but attains only 0.35 estimated utility, while two lower-frequency candidates are executed perfectly. Utilities are estimated from $R = 32$ rollouts per candidate; the aggregate probe, rather than this extreme case, supports the population-level claim.

WebShop reader	score	success@1	success@4
Memory from a DeepSeek teacher			
BASE	0.435	15.9	24.2
MAP	0.394	15.0	24.2
LIST	0.423	16.3	26.0
UNIFORM (redraw)	0.450	17.1	28.5
STORED (frequency redraw)	**0.494**	**17.5**	**29.3**
Teacher-as-agent (DeepSeek)	0.504	18.0	35.1
Memory from a GLM-5.2 teacher			
BASE	0.435	15.9	24.2
MAP	0.427	16.0	24.5
LIST	0.456	18.3	28.8
UNIFORM (redraw)	0.517	20.4	31.9
STORED (frequency redraw)	**0.535**	**21.2**	**33.2**
Teacher-as-agent (GLM-5.2)	0.549	27.3	41.6

Table 3: WebShop reader comparison on the human-written 500-task evaluation split ($n = 4$ attempts). Frequency-weighted STORED is best in both panels and improves over uniform redrawing by 0.8 and 1.3 points at success@4. Memory-construction data, development-time model selection, and the final evaluation split are specified separately in the supplementary material.

ders, and local query strings. Redrawing still prevents all retries from committing to one brittle string, but semantic abstraction would provide a more transferable support for the reader to exploit.

Robustness and cost. The ordering reproduces on ALFWorld `valid_seen`: STORED reaches 74.3 success@8 versus 60.0 for MAP and 61.4 for List. It also reproduces with the independently distilled GLM memory, where STORED reaches 82.1 versus 72.4 and 73.9. All matched readers reuse the same offline teacher bank. STORED introduces no additional test-time model calls relative to MAP or Sample-Once and requires no parameter update. Teacher generation, abstraction, clustering, and storage costs are reported in the supplementary material.

7 Conclusion

Teacher-distilled content raises single-attempt success; STORED adds further gains by sampling the stored distribution instead of collapsing it to a mode, and the readout decomposition prices the two moves separately. Replacing the mode by the marginal, then reopening that marginal across retries. On ALFWorld, STORED reaches 49.0 success@1 and 79.9 success@8 ($+2.5$ and $+11.9$ over MAP), with the redraw gain growing to 9.6 points at thirty-two attempts and the ordering replicating under an independently distilled GLM memory. On WebShop, the same reader is best under both teacher memories (29.3 and 33.2 success@4, up to $+8.7$ over MAP), so the readout ordering transfers across benchmarks, teachers, and guidance representations. Throughout, frequency-weighted redrawing beats uniform redrawing in all four teacher–environment panels, while the utility probe shows why the retained tail matters: the teacher's mode is not the student's best instruction in 40% of probed entries. All of this is achieved with a frozen student, no test-time teacher calls, and no parameter updates. The stored frequencies are an outcome-free prior; letting the agent's own successes and failures re-weight them online is the natural next step. The rule is simple: store the teacher-supported distribution, give the agent one coherent guidance per attempt, and let failure draw again.

Discussion. STORED separates coherence from diversity: within an attempt the frozen student receives one clear instruction; across attempts, failure reopens the memory decision and exposes another teacher-supported direction: independent redrawing being the stateless coupling that preserves the stored marginal and requires neither credit assignment nor utility estimates. Teacher frequency and the distributional tail play complementary roles: frequency wins all four matched comparisons with uniform as an allocation prior, while the oracle probe shows why the tail must be retained: lower-frequency recommendations can be substantially easier for the frozen student to execute. The finite-bank view ties the two budgets together: collection fixes which alternatives can ever be exposed, retries only explore that fixed bank, and semantic candidate count matters only through the utility heterogeneity it induces, captured by $\mathrm{Var}(U_x)$.

Limitations. The retry decomposition conditions on the complete attempt-level readout map and assumes fresh residual randomness across attempts, modelling the reset-style success@k protocol but not state carried across retries. The exchangeable-slot model abstracts the data-dependent aggregation and filtering, and N_i is not varied experimentally, so monotonicity in bank size remains a design prediction. WebShop evaluates the complete reader without a matched Sample-Once control and stores literal rather than abstracted guidance. The utility probe covers 70 entries and is diagnostic only.

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